GS EXTERIOR EXPERTS · DECK DIVISION · PERMIT SET

281 S Jefferson Ave · Decks + Roof Covers

281 S Jefferson Ave, Louisville · two decks (12' x 12' + 14'8" x 3'8" walkway) on CEMCO steel and Diamond Piers · TimberTech Landmark American Walnut, picture frame + dividers, no butt joints · two lean-to covers, black standing seam, Hardie soffit / fascia / rakes · blinds kept · October 16, 2026

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15. Decks + roof covers · **D-503** Connection details — roof covers

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. This set carries no pricing.

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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PROJECT

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel, TimberTech Landmark American Walnut)
281 S Jefferson Ave, Louisville CO 80027
197.8 SF · 1'-3" above grade · 2 zones

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; ground snow per the ASCE hazard tool (Louisville criteria) — 30 psf used here (city sheet, non-reducible); Boulder County layer shows 40 psf: engineer confirms
Dead load	10 psf deck · 15 psf porch cover (standing seam steel on sheathing, Hardie soffit; 12 psf actual, 15 used) carried on the deck posts
Roof snow (cover)	30 psf flat-roof design snow, sliding snow off the house roof + drift at the house wall per ASCE 7 Ch. 7
Design total	50 psf on the deck · cover posts add roof snow + dead to the deck frame and footings (engineered)
Wind	145 mph Vult (3-second gust), Exposure C (Louisville 2024 design criteria), Risk Cat. II
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2024 IRC / IBC as adopted and amended by the City of Louisville (Ord. 1907, Series 2025); IRC R507 decks with LMC 15.05.140; IBC Appendix I patio covers; ASCE 7-22 loads; Louisville Fire Hardening provisions; cold-formed steel frame per AISI S240, engineered
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	3 x 6x6 Douglas fir rough deck posts
Beams	FLUSH boxed 600S162-54 W: 14'-1 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 471 lb); FLUSH boxed 600S162-54 M: 11'-5 1/2", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst 1,613
Joists	CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC
Rims	single-ply 600T track rims (G90) and 600T ledger track on a steel angle
Ledger	600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
Decking	TimberTech Landmark American Walnut — 25 rows parallel to the house, picture frame & dividers in Landmark American Walnut, 3/16" gaps, CONCEALoc
Rail	none

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SITE

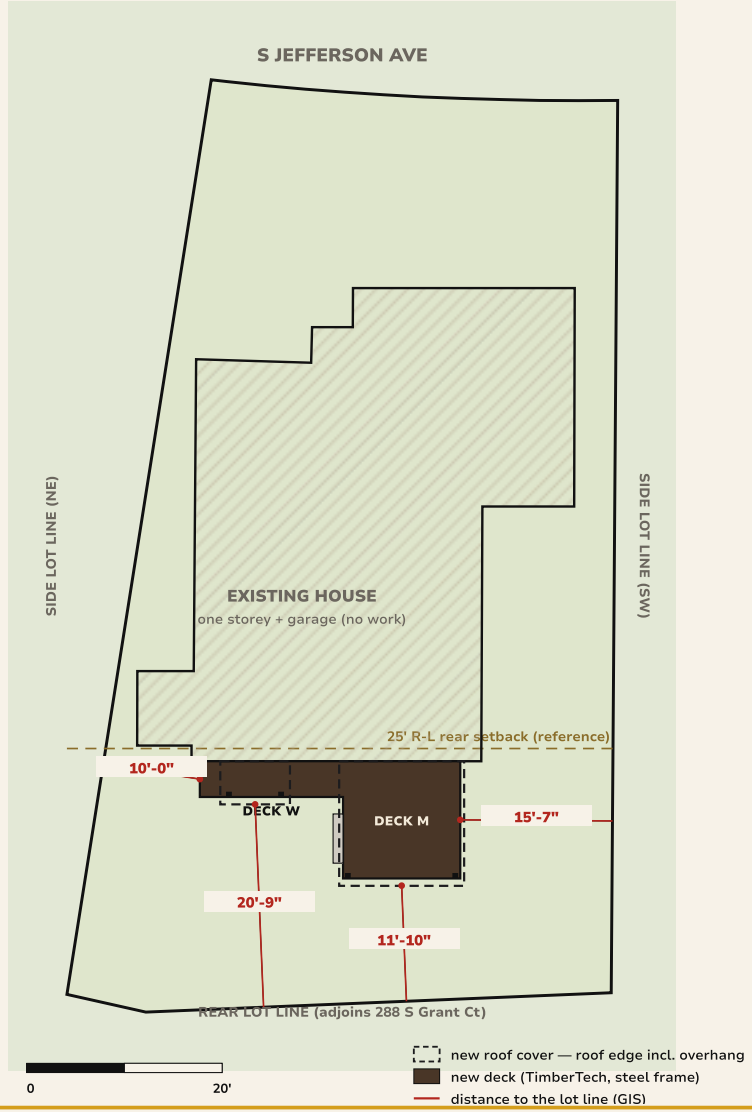
Rear / front lot line: deck edge 12.0' from the line, setback per the PUD...
Side lot line: deck edge 9.8' from the line, setback per the PUD —...

C-100 SITE PLAN

Parcel, existing house, new decks and roof covers, distances to the lot lines, coverage

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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PARCEL

281 S Jefferson Ave, Louisville CO 80027
Parcel 157517119033 · Account R0092293
Lot 33, Block 10, Dutch Creek (plat rec. 00492001, 1982)
Section 17, T1S, R69W
Zoning: R-L within the Dutch Creek PUD (approved 12/15/1981)
Lot 4,678 SF · flood zone X
House 1982 one-storey ranch: 920 SF main floor + 400 SF garage (assessor)

LOT COVERAGE

Lot area	4,678 SF
House + garage (assessor footprint)	1,320 SF
Existing covers to the posts (2024 aerial, est.)	~142 SF (main ~12' x 10'-4", walkway ~5'-4" x 3'-5")
Existing coverage	~1,462 SF · 31.3%
Main cover, new (to the posts) 12'-0" x 11'-9"	141 SF
Walkway cover, new (to the posts) 5'-4" x 3'-5"	18 SF
Proposed coverage	~1,479 SF · 31.6% (+~17 SF)
Uncovered decks	not counted (under 30" above grade)

SETBACKS

Decks are 30" or less above grade — not held to the principal setbacks (city deck handout)
Roof covers: part of the principal structure — they replace the two existing covers in place; the 2024 aerial shows the existing main roof ~1'-5" shallower: field-verify the existing post line
R-L reference setbacks (LMC 17.12.040) — the Dutch Creek PUD plan governs; Planning to confirm
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the two decks, the two roof covers and the box step; haul off
· Diamond Pier foundations (no excavation spoils), 6-mil poly + 2" gravel under the decks
· Stair lands on grade (no pad)
· Roof drainage: covers shed to grade at the low eave, away from the house
· Call 811 before the piers; protect the lawn and the owner's blinds

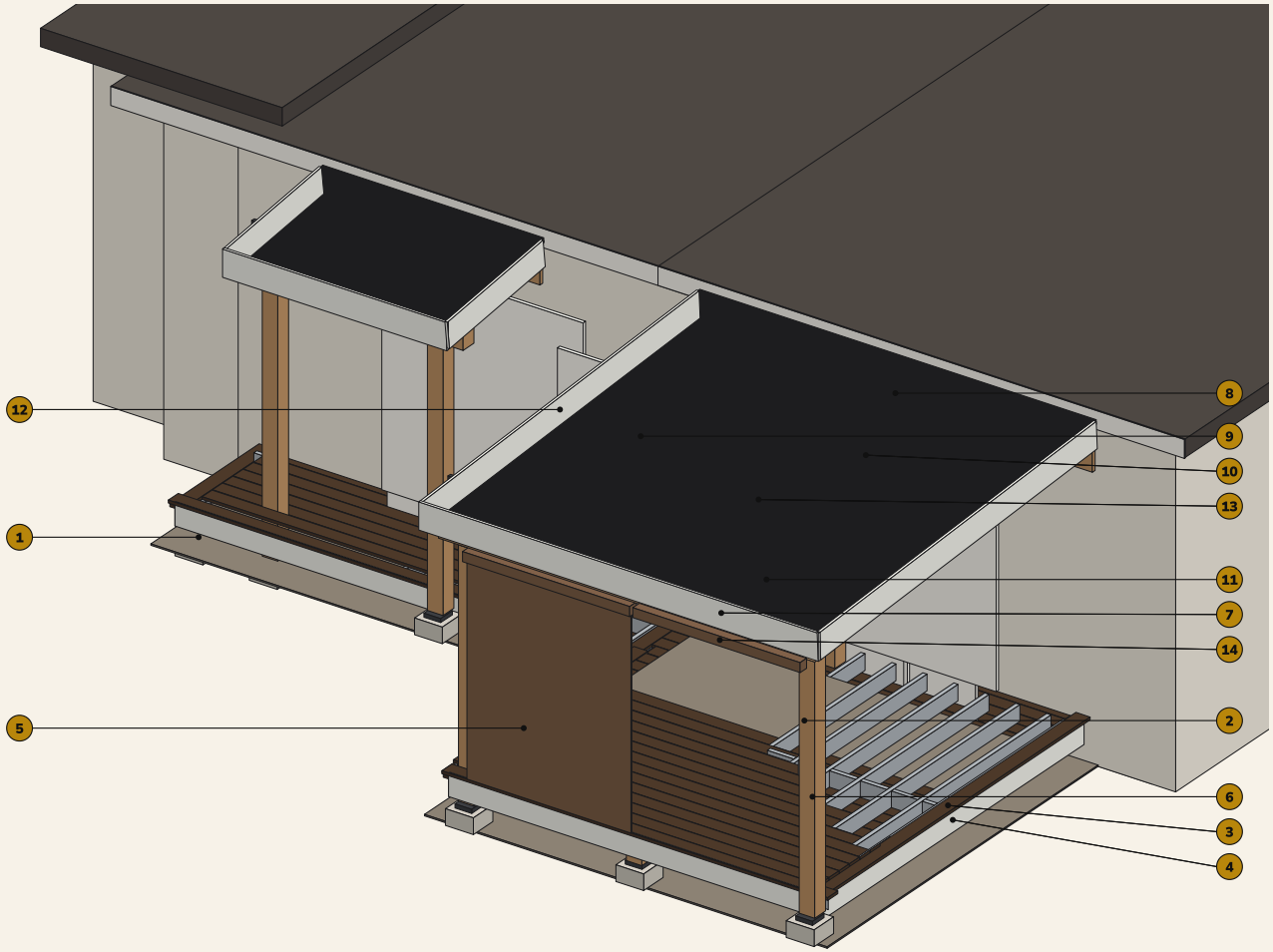
Boulder County GIS parcel + 2024 Pictometry roof outline, Louisville GIS zoning / PUD layers. GIS lines are not a survey — an ILC governs if Planning requires one.

A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. TimberTech Landmark American Walnut field decking, hidden fasteners
3. Picture frame + breaker boards — TimberTech Landmark American Walnut
4. HardieTrim skirt Arctic White
5. Stair — steel stringers, Landmark treads, closed risers, lands on grade
6. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
7. 4x10 cover beam, Douglas fir, BC46Z cap
8. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
9. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
10. Black standing seam, 24 ga, mechanically seamed
11. Snow retention bar, clamped to the seams
12. HardieTrim fascia + rakes Arctic White
13. HardieSoffit cedarmill ceiling Arctic White
14. Existing roll-down blinds (stay)

House shown for context only (no work). The CEMCO steel deck frame under the decking is called out on S-102 and A-003; cover framing is wood: 2x8 rafters, main cover 4x10 beam, walkway cover (2)2x8 beam, 6x6 Douglas fir posts.

Axonometric · not to scale · the model the plans are drawn from

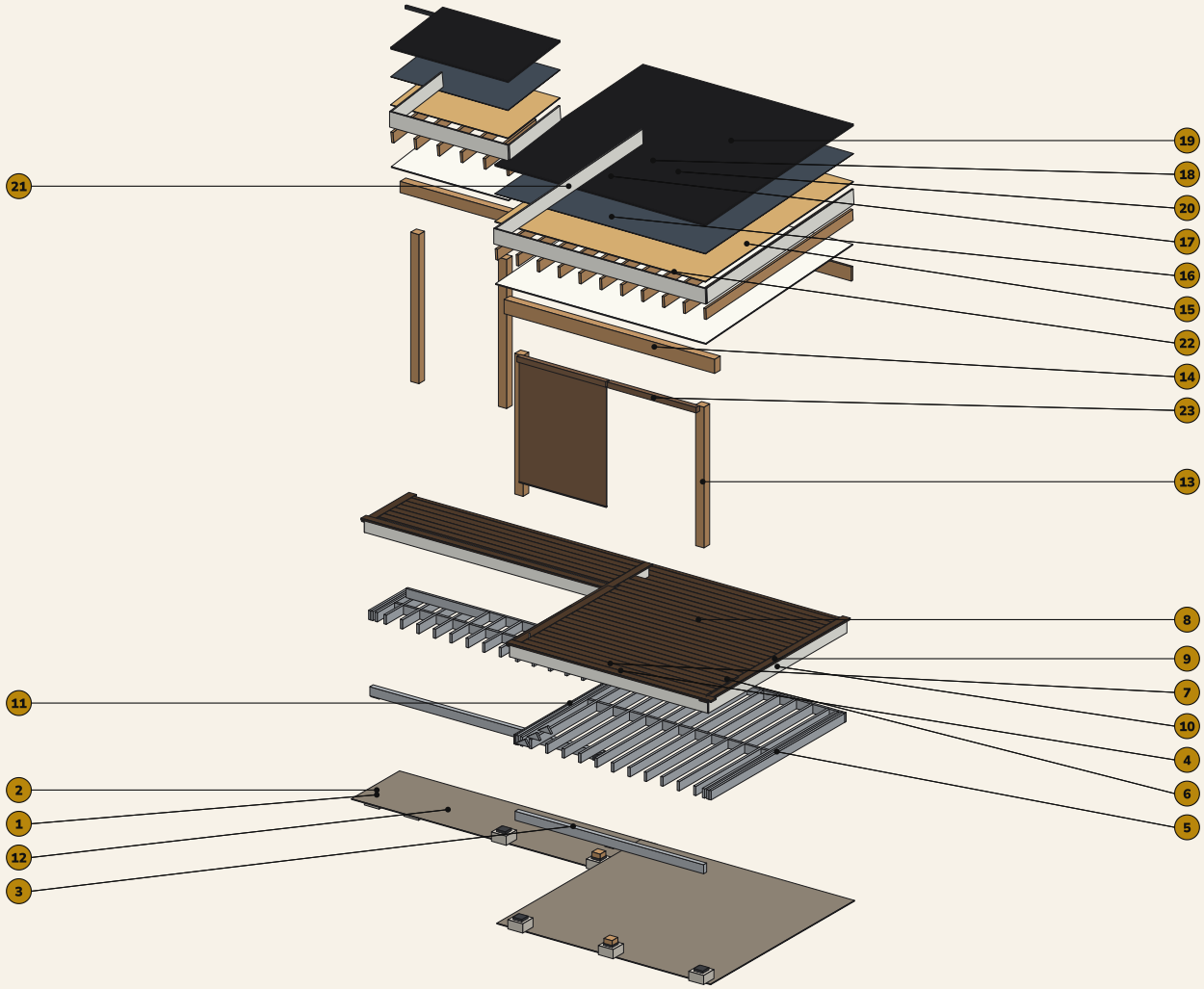
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-003 ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts · beams · frame · decking · stair · roof covers layer by layer

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir rough deck post on ABA66RZ, steel saddle to the boxed beam
3. Boxed CEMCO 600S162-54 beam (2 C toe-to-toe, G90), flush
4. CEMCO 600S162-54 C-joist, G90, @ 12" OC — clip angles each end
5. CEMCO 600T track rim, G90 (caps the joist ends)
6. Steel ledger track + 3x3 ledger angle on the house rim, flashed
7. Steel blocking at the breaker / picture-frame joists
8. TimberTech Landmark American Walnut field decking, hidden fasteners
9. Picture frame + breaker boards — TimberTech Landmark American Walnut
10. HardieTrim skirt Arctic White
11. Stair — steel stringers, Landmark treads, closed risers, lands on grade
12. Gravel over 6-mil poly under the decks
13. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
14. 4x10 cover beam, Douglas fir, BC46Z cap
15. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
16. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
17. 7/16" OSB roof sheathing
18. High-temp self-adhered underlayment
19. Black standing seam, 24 ga, mechanically seamed
20. Snow retention bar, clamped to the seams
21. HardieTrim fascia + rakes Arctic White
22. HardieSoffit cedarmill ceiling Arctic White
23. Existing roll-down blinds (stay)

Layers lifted in build order. Roof covers pulled apart: posts, beam + ledger, rafters, soffit, sheathing, underlayment, standing seam, snow bar, Hardie trim.

Axonometric · not to scale · the model the plans are drawn from

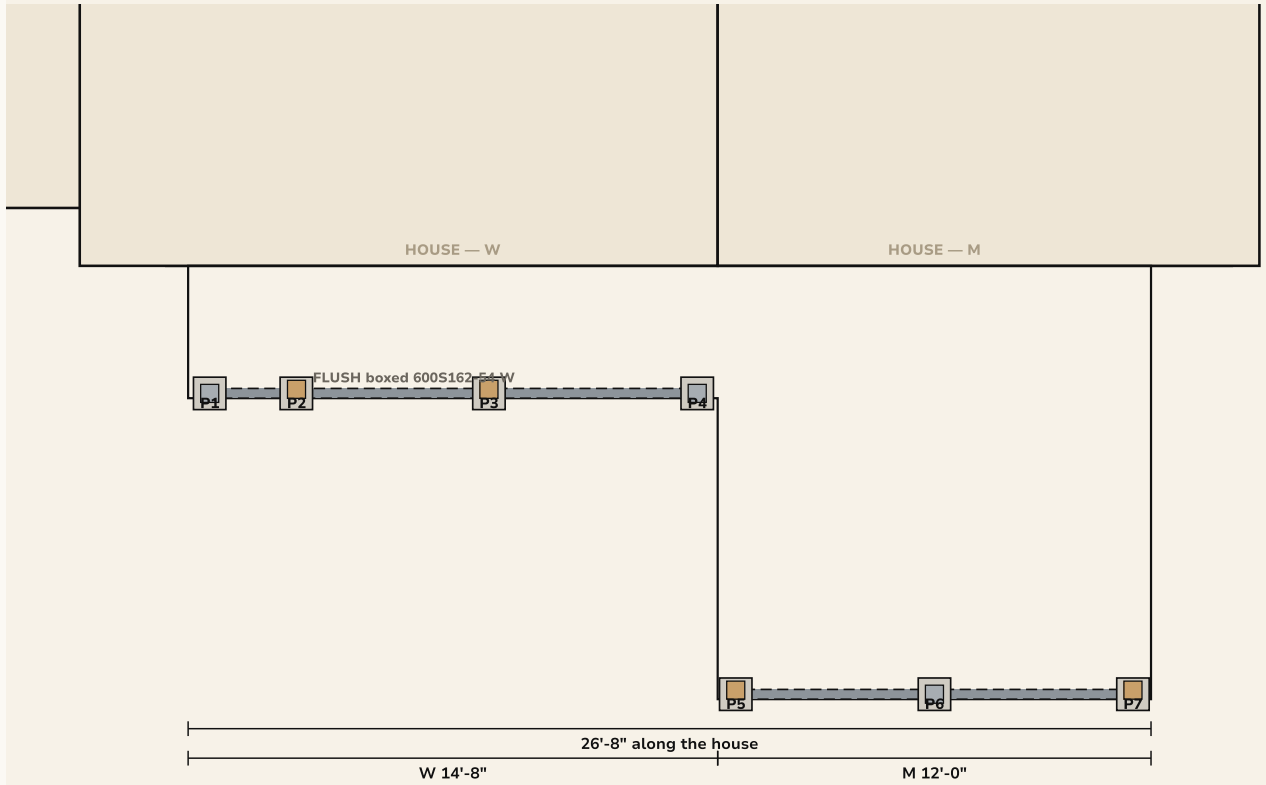
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins),
3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	7 3/16"	3'-6 3/8"	W	130 lb
P2	3'-0"	3'-6 3/8"	W +cover	328 lb +roof
P3	8'-4"	3'-6 3/8"	W +cover	471 lb +roof
P4	14'-1 3/16"	3'-6 3/8"	W	270 lb
P5	15'-2"	11'-10 3/8"	M +cover	874 lb +roof
P6	20'-8"	11'-10 3/8"	M	1,613 lb
P7	26'-2"	11'-10 3/8"	M +cover	873 lb +roof

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,613 lb

Deck-only piers: 6x6 Douglas fir rough post on ABA66RZ, steel saddle to the bea
+cover: the 6x6 Douglas fir rough roof-cover post stands on this pier
(ABA66RZ) and carries the deck beam on a steel side bracket (D-502 / 4);
roof load adds — the engineer sizes the pier.
Beams: boxed CEMCO 600S162-54, flush (D-502 / 2).

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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FRAMING NOTES

CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC

single-ply 600T track rims (G90) and 600T ledger track on a steel angle

600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer

blocking one row over the beam, staggered

FLUSH boxed 600S162-54 W: 14'-1 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 471 lb); FLUSH boxed

Connectors:

Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam: 56

Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2

Cold-galvanizing spray (cut ends, field drilling): 2

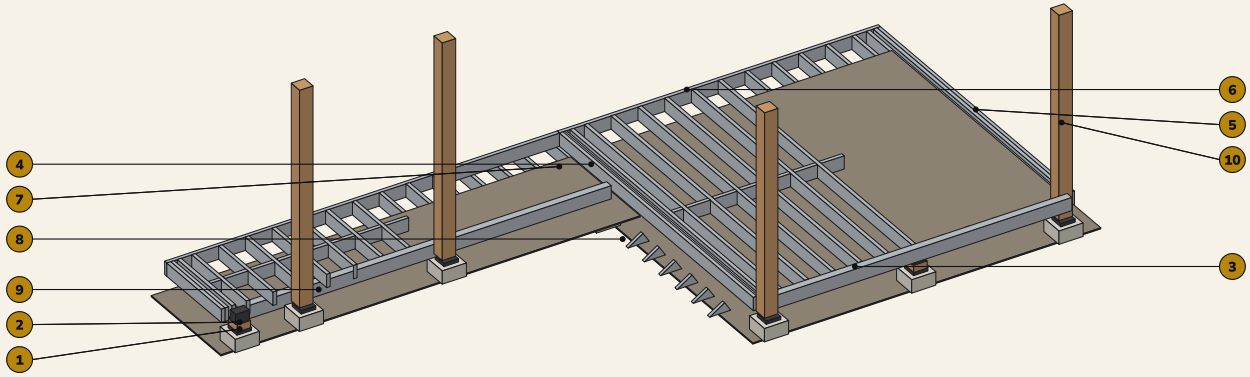
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-102 STRUCTURE ISOMETRIC — STEEL FRAME, LABELED

Piers, stubs, cover posts at the shared piers, boxed beams, C-joists, track rims, ledger track, blocking

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir rough deck post on ABA66RZ, steel saddle to the boxed beam
3. Boxed CEMCO 600S162-54 beam (2 C toe-to-toe, G90), flush
4. CEMCO 600S162-54 C-joist, G90, @ 12" OC — clip angles each end
5. CEMCO 600T track rim, G90 (caps the joist ends)
6. Steel ledger track + 3x3 ledger angle on the house rim, flashed
7. Steel blocking at the breaker / picture-frame joists
8. Stair — steel stringers, Landmark treads, closed risers, lands on grade
9. Gravel over 6-mil poly under the decks
10. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)

Decking, covers and house removed to show the frame. Every joist end clips to the track, ledger track or boxed beam; the engineer's clip schedule governs.

A-101 DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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DECKING

Field: TimberTech Landmark American Walnut, 5.5" x 1", 3/16" gaps

Border & dividers: Landmark American Walnut

Boards run parallel to the house; edge overhang 1" over the fascia

CONCEALoc at every joist in every gap (437)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per panel: panel 1 (0"–14'-8") 7 rows · panel 2 (14'-8"–26'-8") 25 rows · 2 ripped pieces · 1 breaker — every board one piece wall to border, no butt

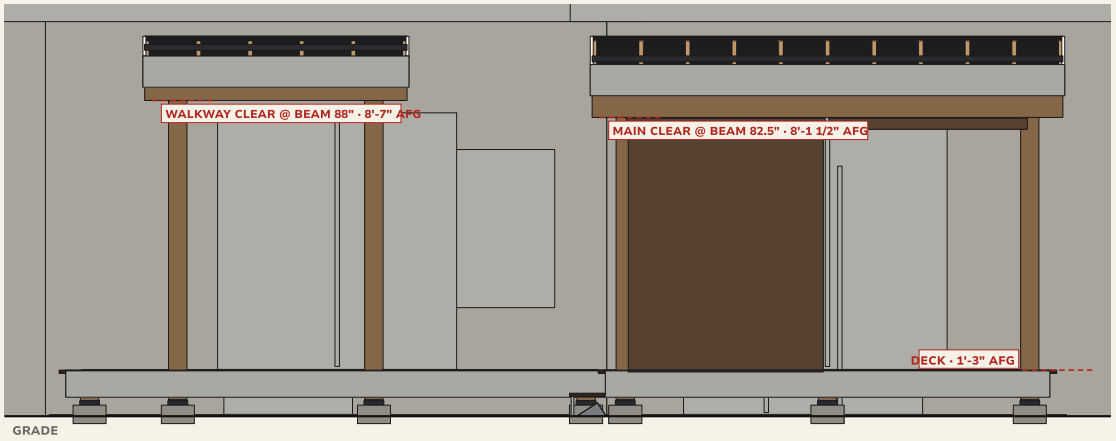
A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

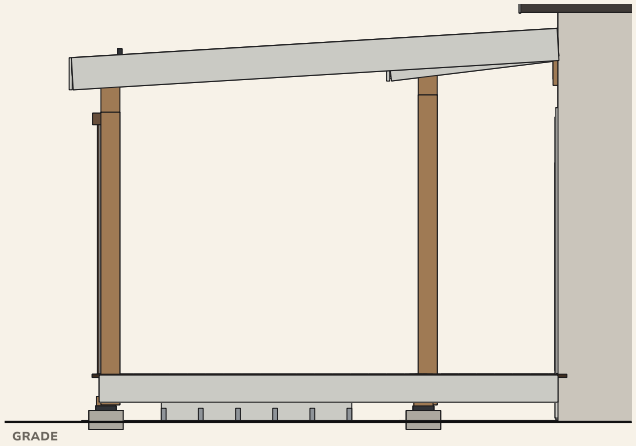
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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FRONT ELEVATION — from the yard



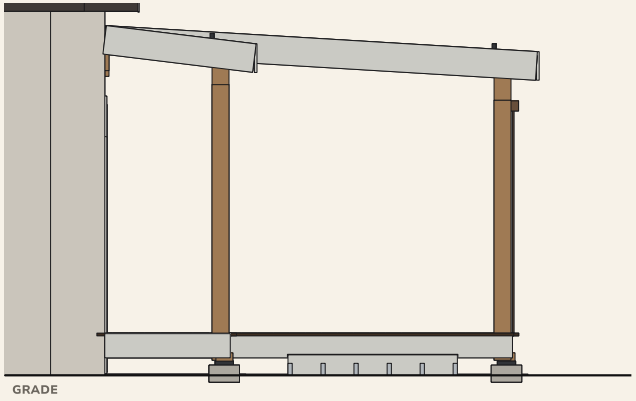
RIGHT END ELEVATION



NOTES

· Deck surface 1'-3" above grade (walkway low point 9")
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Clear heights are to the underside of the beam / ledger above the deck surface
· Roof covers: black standing seam on 2x8 rafters, Hardie fascia / rakes / soffit; slope to the yard
· Deck frame: CEMCO cold-formed steel; HardieTrim skirt covers the track rims
· No guard required: every walking surface is under 30" above grade (IRC R312.1)

LEFT END ELEVATION



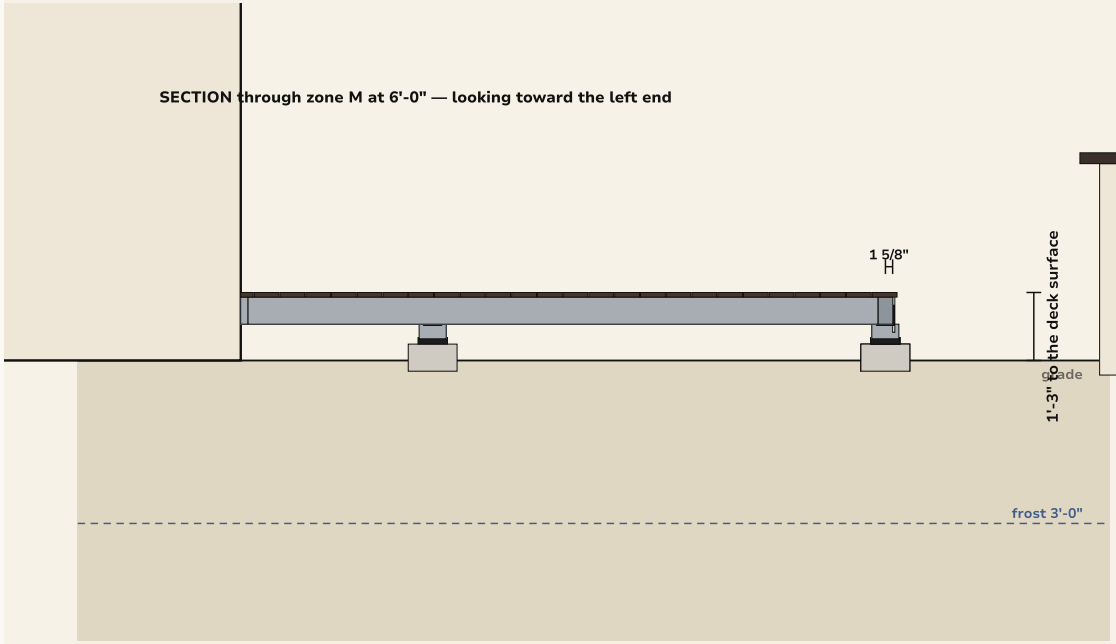
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

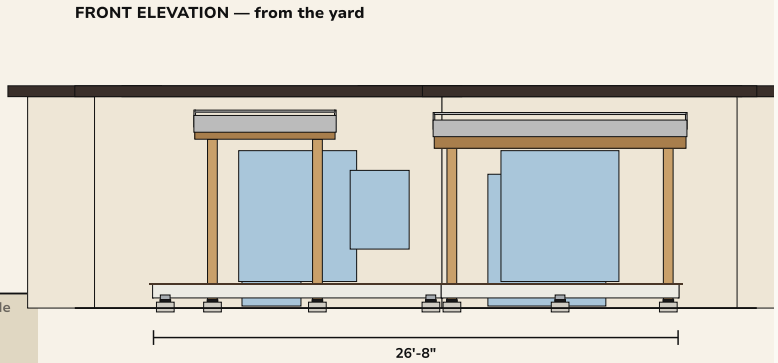
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

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SECTION NOTES

Deck surface 1'-3" above grade

FLUSH boxed 600S162-54 W: 14'-1 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 471 lb); FLUSH boxed 600S162-54 M: 11'-5 1/2", 3 posts at 15'-2" / 20'-8" / 26'-2"

Posts 3 x 6x6 Douglas fir rough deck posts on steel beam saddles + 4 6x6 Douglas fir rough cover posts carrying the deck beam on DP-50/50

600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

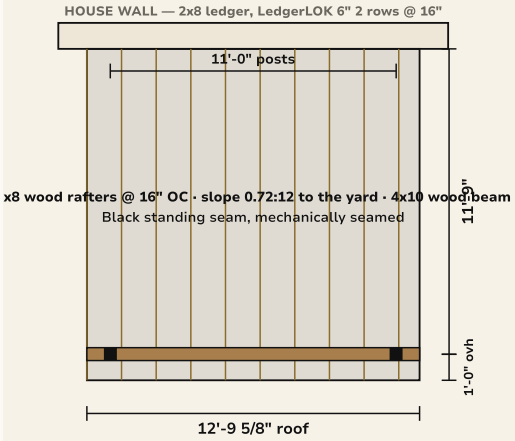
A-401 ROOF COVERS — PLAN + SECTION

Lean-to covers: ledger, rafters, beam, posts on the piers, clear heights, pitch, standing seam, Hardie, blinds

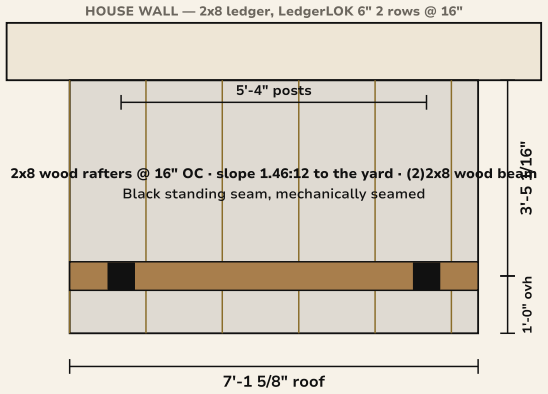
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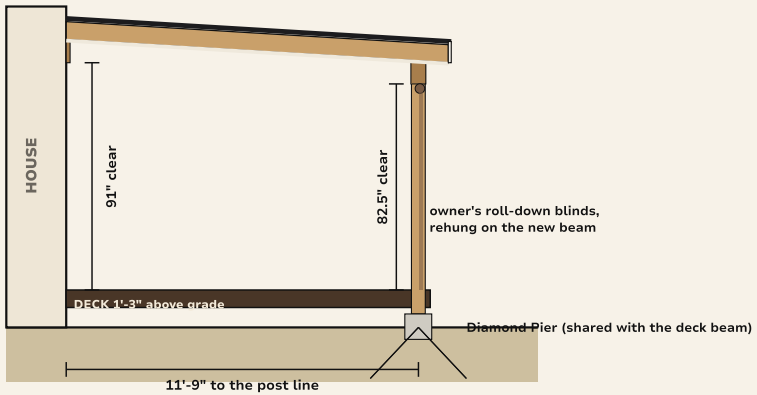
MAIN COVER — PLAN



WALKWAY COVER — PLAN

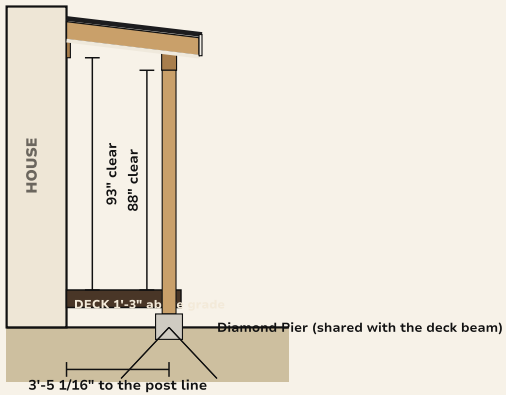


MAIN COVER — SECTION



Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 0.72:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · post caps per D-503
6x6 Douglas fir rough posts on ABA66RZ bases on the Diamond Pier bolt · BC46Z cap to the 4x10 Douglas fir beam · the engineer's uplift hold-down governs

WALKWAY COVER — SECTION



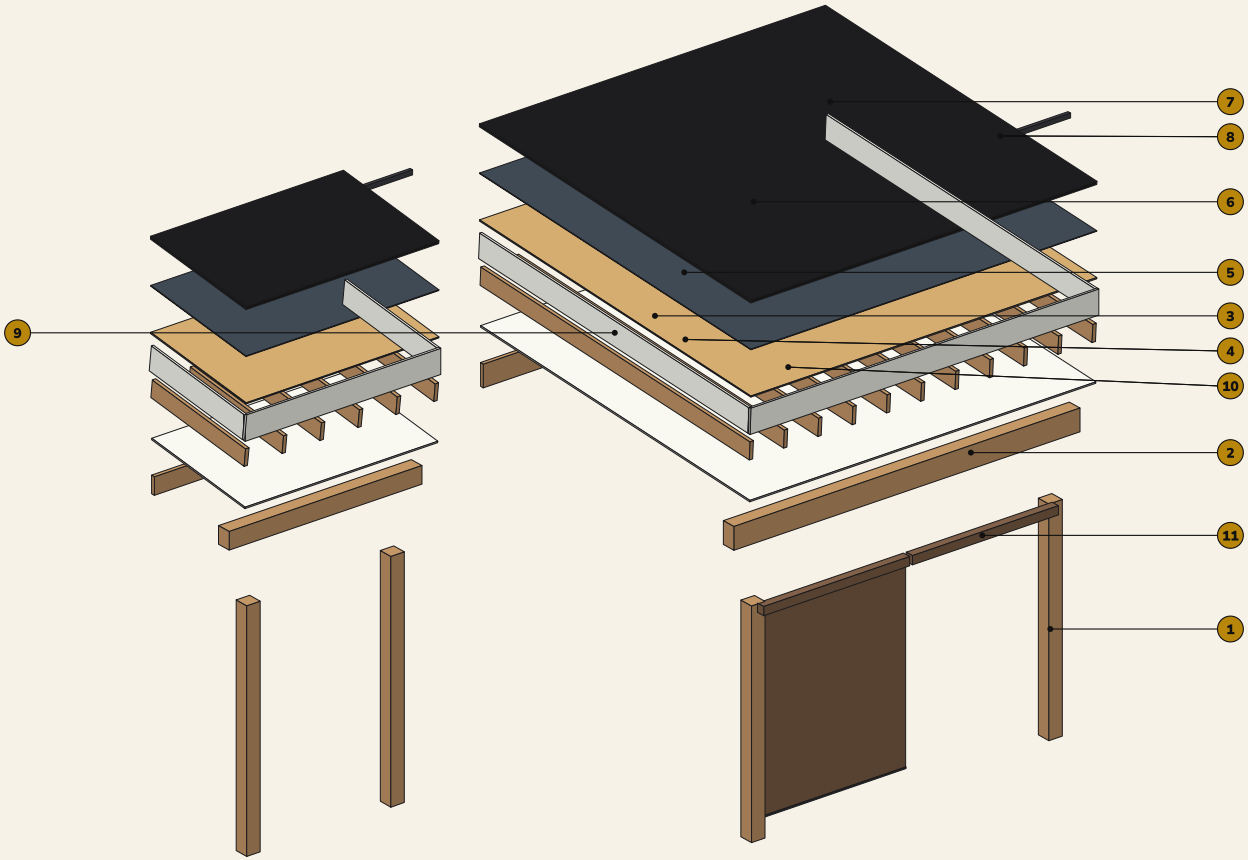
Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 1.46:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · post caps per D-503
6x6 Douglas fir rough posts on ABA66RZ bases on the Diamond Pier bolt · AC6Z cap to the (2)2x8 beam · the engineer's uplift hold-down governs

A-402 ROOF COVERS — EXPLODED ISOMETRIC, LABELED

Both lean-to covers pulled apart layer by layer

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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KEY

1. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
2. 4x10 cover beam, Douglas fir, BC46Z cap
3. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
4. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
5. 7/16" OSB roof sheathing
6. High-temp self-adhered underlayment
7. Black standing seam, 24 ga, mechanically seamed
8. Snow retention bar, clamped to the seams
9. HardieTrim fascia + rakes Arctic White
10. HardieSoffit cedarmill ceiling Arctic White
11. Existing roll-down blinds (stay)

Lifts are for clarity only. Order of work: posts on ABA66RZ bases, beams on their post caps, ledger on LedgerLOK, rafters with H2.5AZ, soffit, OSB, underlayment, panels, trims, snow bar.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

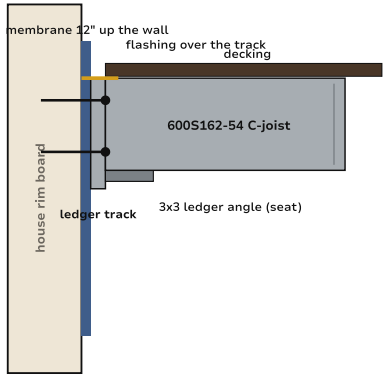
D-501 DETAILS — STEEL FRAME

Steel ledger · joist clips · boxed beam on the shared pier · Diamond Pier · deck edge · under the deck

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

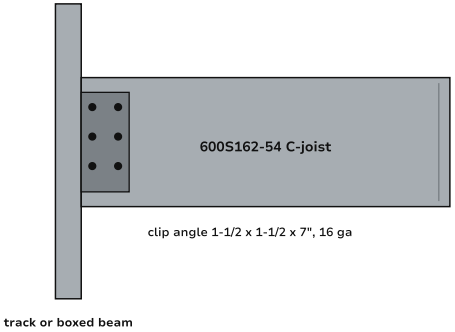
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1 STEEL LEDGER AT THE HOUSE



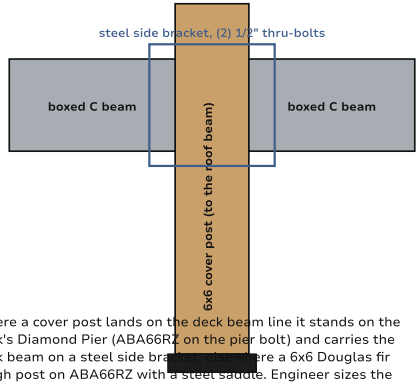
600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

2 JOIST TO TRACK / BEAM



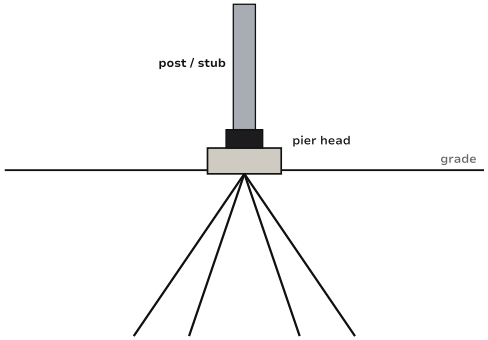
#12-14 self-drilling hex-washer screws, 4 per leg minimum (clip to joist web, clip to track / beam web) — the engineer's clip schedule governs. Cut ends and field holes cold-galvanized.

3 BOXED BEAM · SHARED PIER



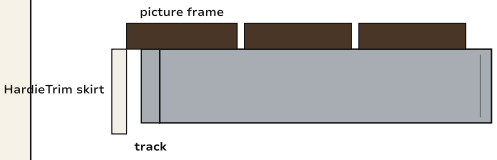
Where a cover post lands on the deck beam line it stands on the deck's Diamond Pier (ABA66RZ on the pier bolt) and carries the deck beam on a steel side bra[illegible]re a 6x6 Douglas fir rough post on ABA66RZ with a steel saddle. Engineer sizes the pier for roof + deck and the uplift hold-down.

4 DIAMOND PIER



DP-50/50 (4 pins) — no excavation; frost-protected by the pins (submit the evaluation report). 811 locate first.

5 DECK EDGE · HARDIE SKIRT



TimberTech Landmark American Walnut: hidden fasteners in the field, the picture frame face-fastened with color-matched plugs — every screw the maker's steel-frame screw for 14–18 ga. HardieTrim skirt on furring screwed to the track.

6 UNDER THE DECK · LOW END



2" gravel over 6-mil poly, sloped away from the house

Keep 2" min. between the frame and the gravel.

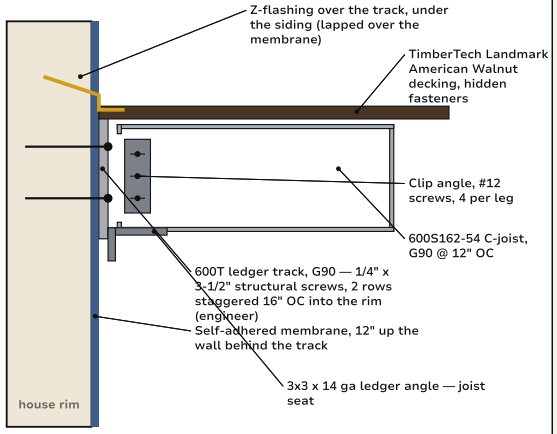
D-502 CONNECTION DETAILS — CEMCO STEEL DECK FRAME

Ledger track · joist clips · boxed beam · post on the pier · shared pier · deck edge · stair

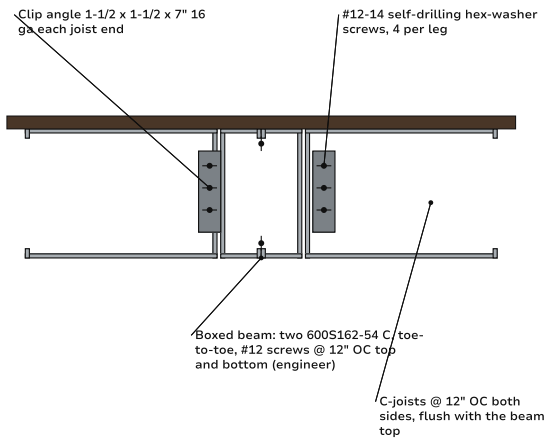
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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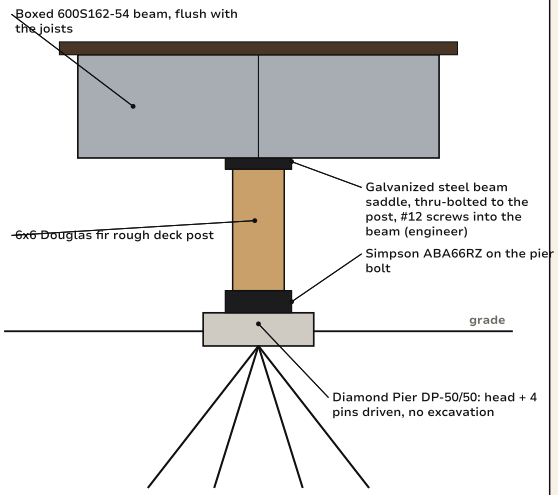
1 LEDGER TRACK AT THE HOUSE — SECTION



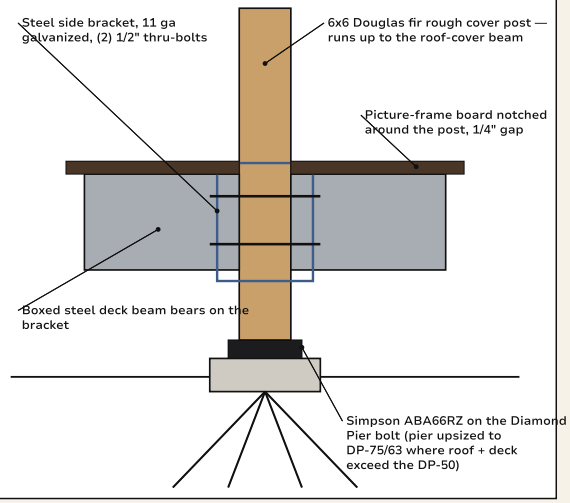
2 JOIST TO BOXED BEAM — SECTION



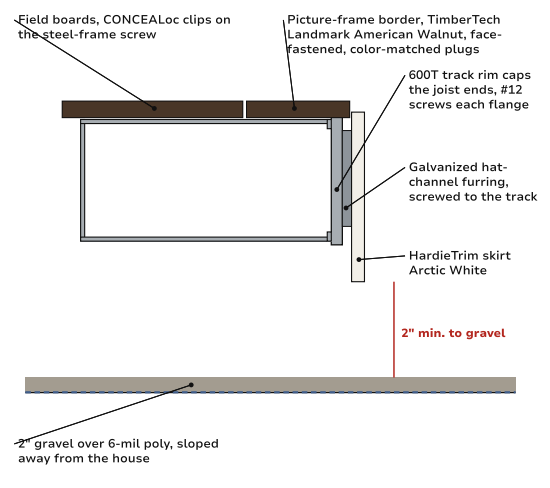
3 BEAM ON 6x6 POST + DIAMOND PIER



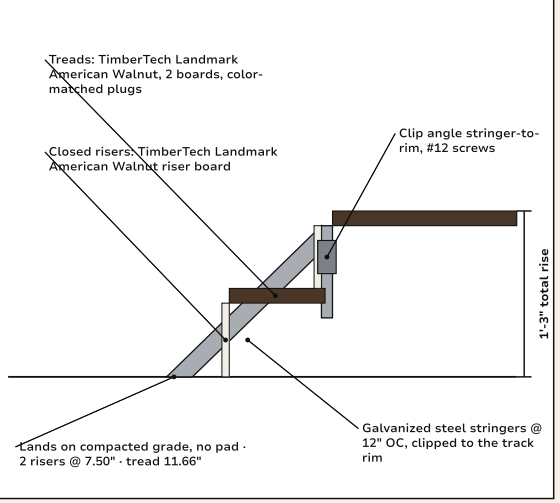
4 SHARED PIER — COVER POST + DECK BEAM



5 DECK EDGE — TRACK RIM, BORDER, HARDIE SKIRT



6 STAIR OFF THE MAIN DECK — SECTION



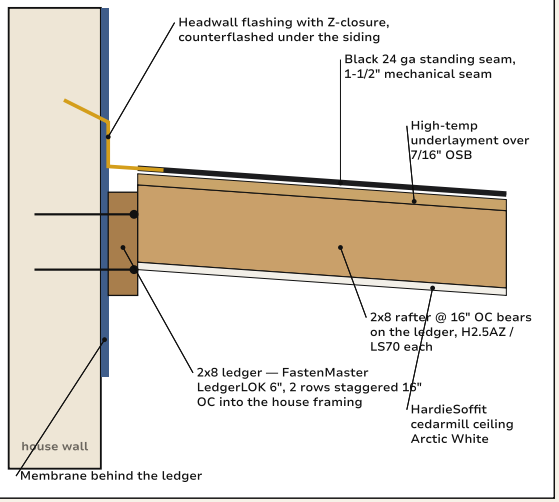
D-503 CONNECTION DETAILS — ROOF COVERS (WOOD FRAMING)

Ledger + headwall · eave at the beam · post base · rake · standing seam + clip · blinds

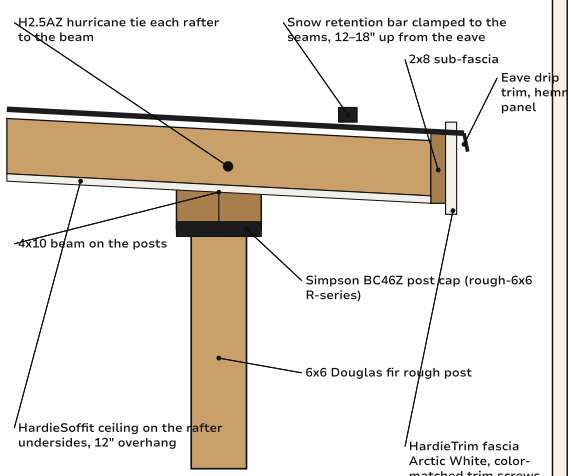
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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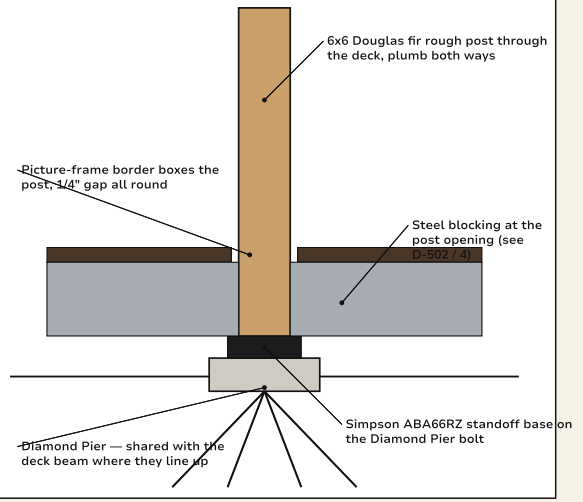
1 COVER LEDGER + HEADWALL — SECTION



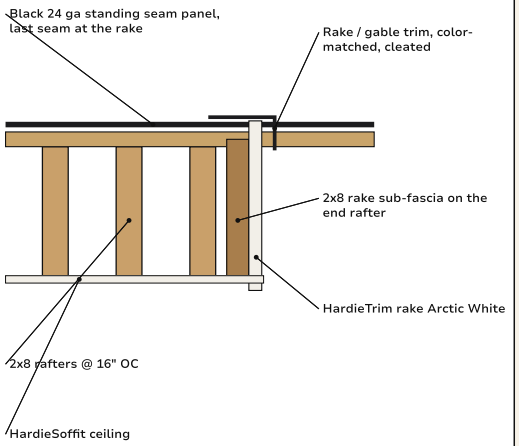
2 EAVE AT THE BEAM — SECTION



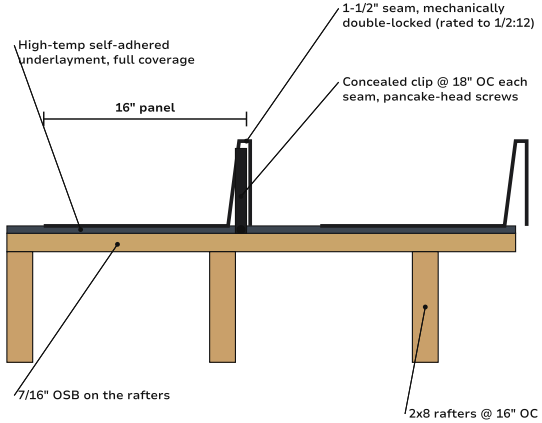
3 COVER POST BASE + DECK PENETRATION



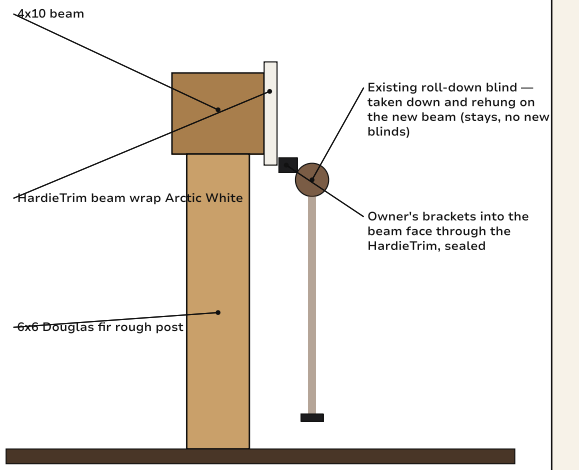
4 RAKE EDGE — SECTION ACROSS THE SLOPE



5 STANDING SEAM PANEL + CLIP



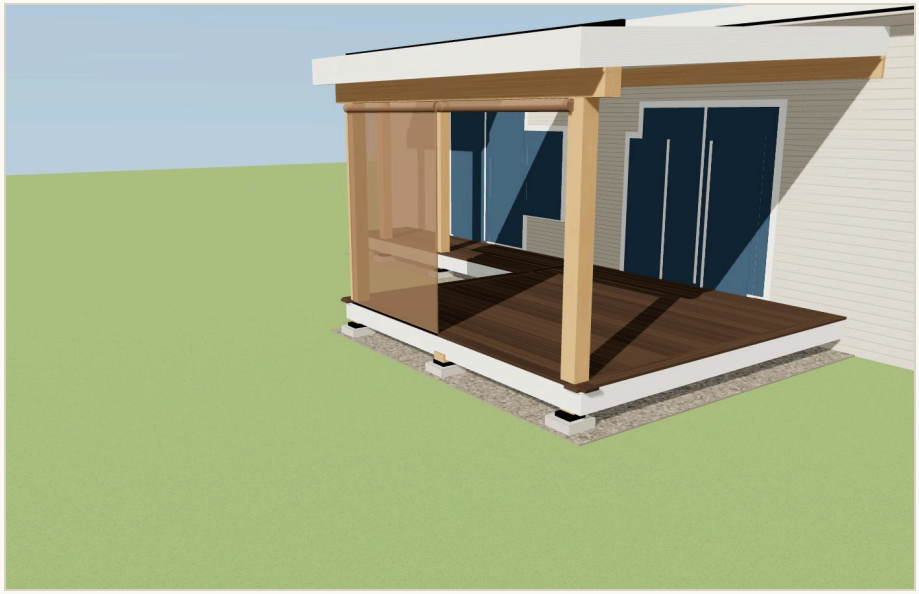
6 EXISTING BLINDS AT THE BEAM



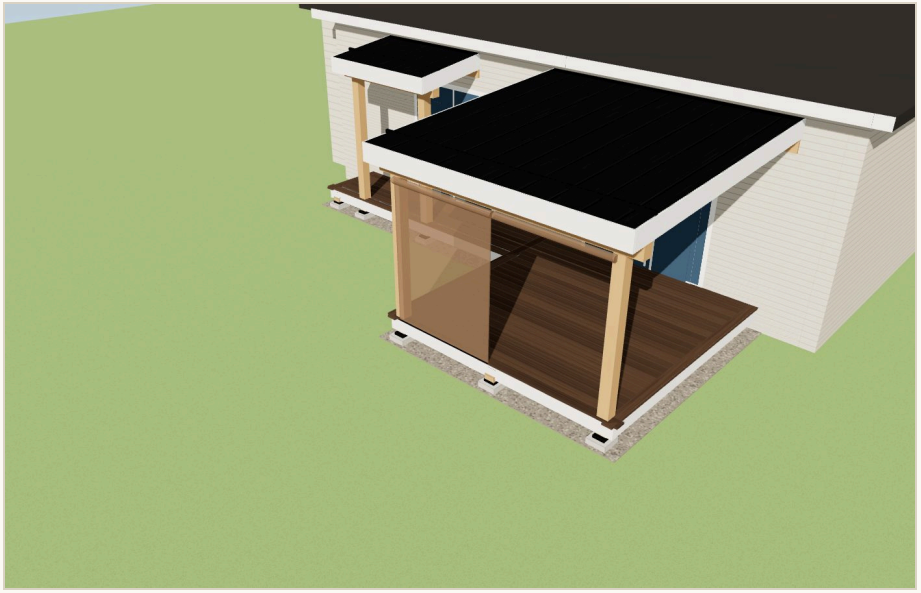
Decks + roof covers



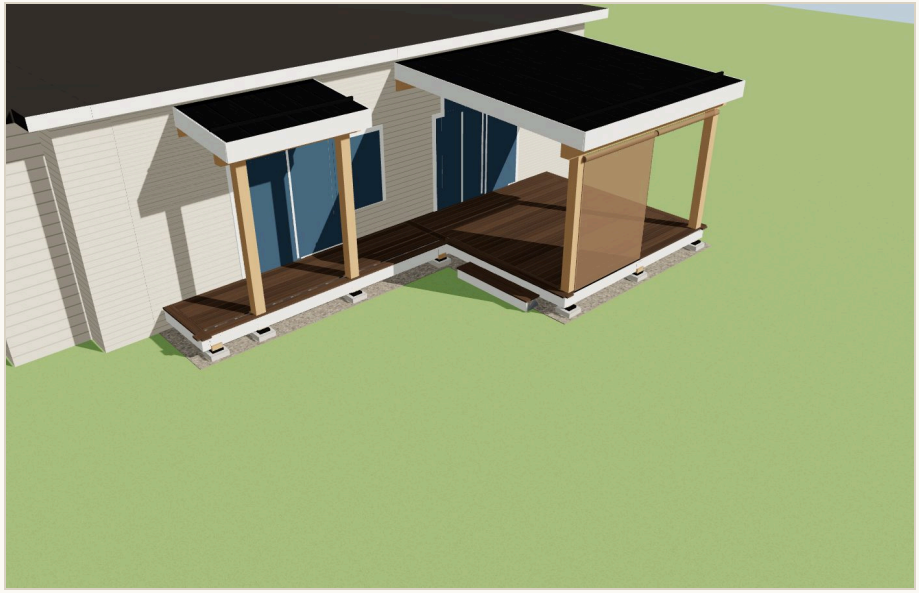
From the yard



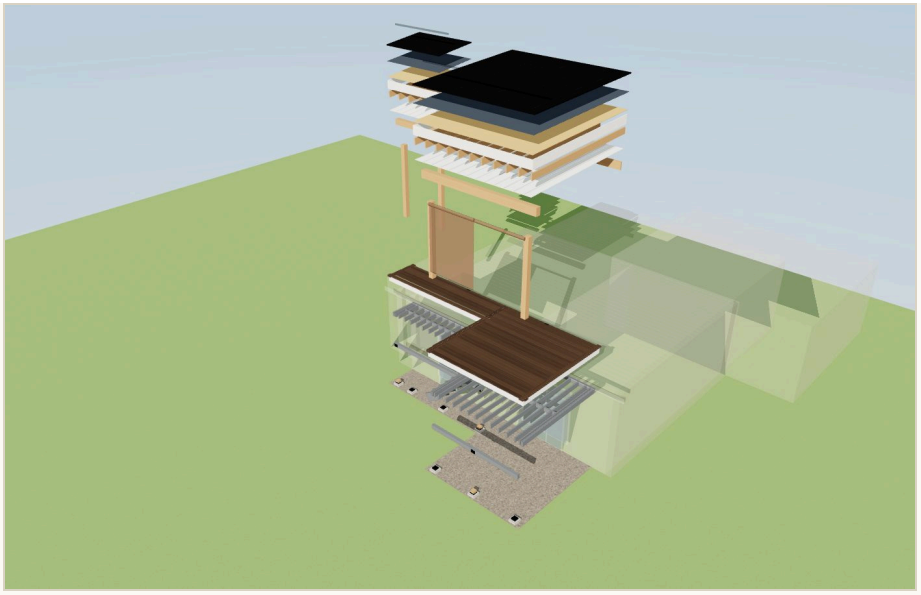
From the front corner



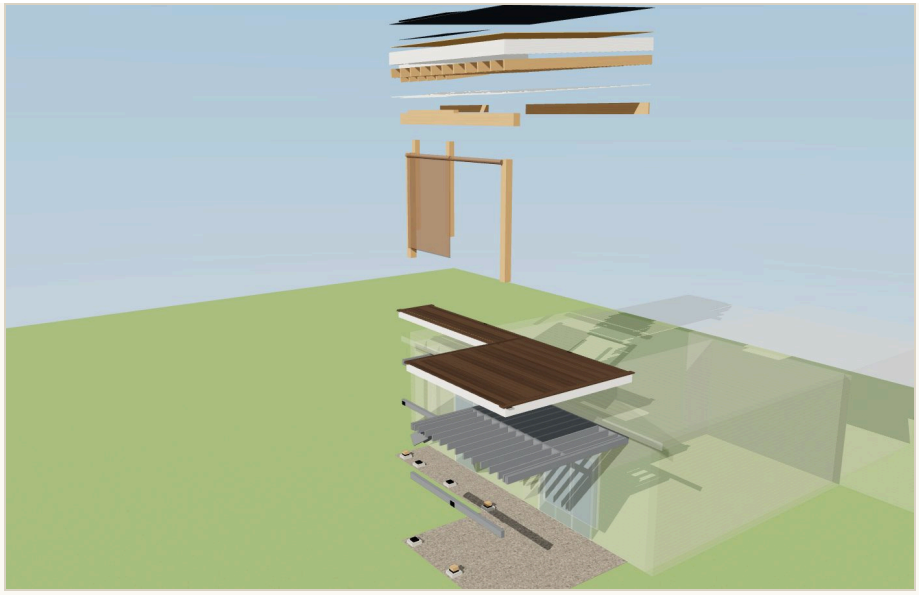
Isometric



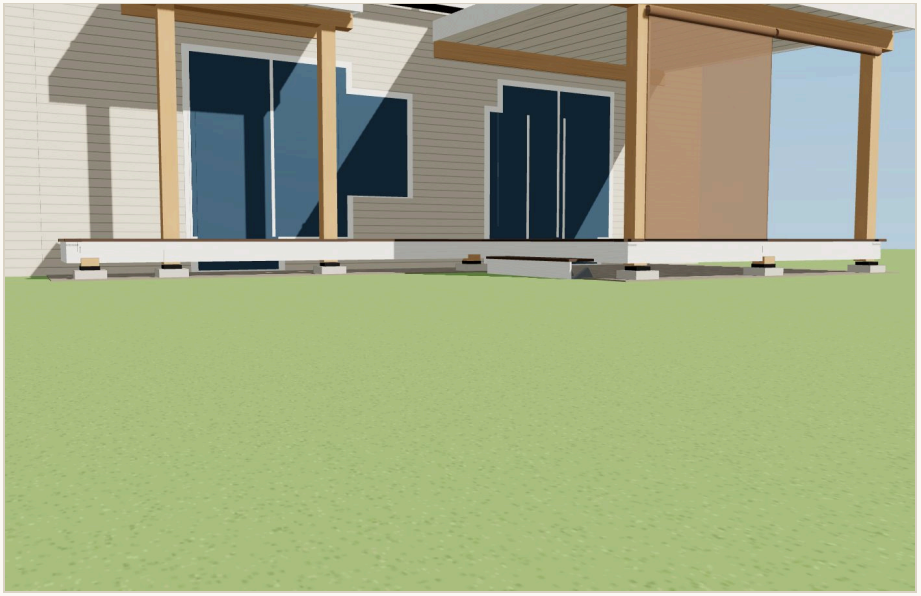
Isometric — opposite corner



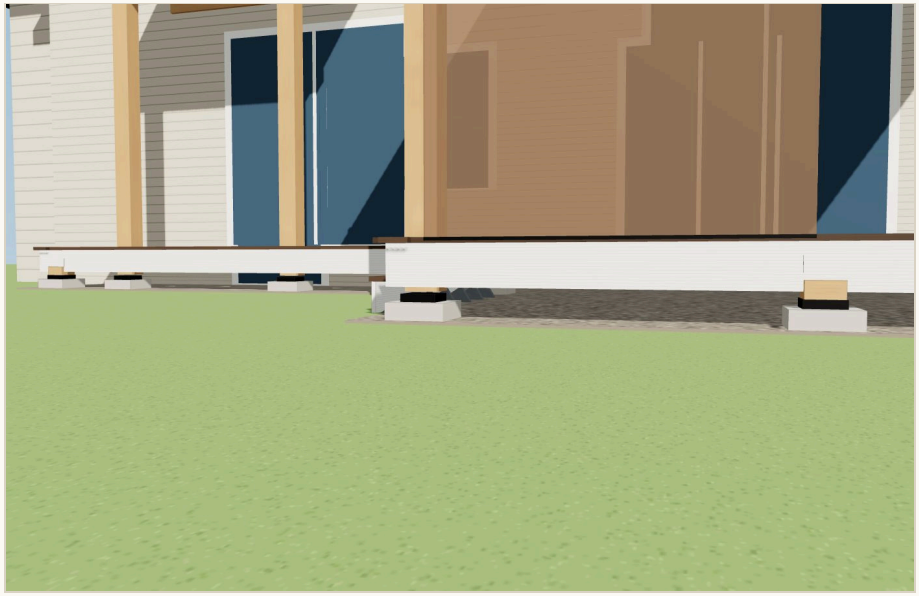
Exploded — every layer, covers included



Exploded — from the corner



Structure from below



Under the deck